# AISI Type 316L Stainless Steel, annealed sheet

**Categories:** Metal; Ferrous Metal; Stainless Steel; T 300 Series Stainless Steel

**Material Notes:** Similar to Type 316 for superior corrosion resistance, but also has superior resistance to intergranular corrosion following welding or stress relieving. Good corrosion resistance to most chemicals, salts, and acids and molybdenum content helps resistance to marine environments. The low carbon content of 316L reduces the possibility of in vivo corrosion for medical implant use. High creep strength at elevated temperatures. 316L has fabrication characteristics similar to Types 302 and 304.

**Applications:** biomedical implants, chemical processing, food processing, photographic, pharmaceutical, textile finishing, marine exterior trim.

**Key Words:** UNS S31603, AISI 316L, ISO 2604-1 F59, ISO 2604-4 P57, ISO 2604-4 P58, ISO 4954 X2CrNiMo17133E, ISO 683/13 19, ISO 683/13 19a, biomaterials, biomedical implants, biocompatible materials

**Vendors:** No vendors are listed for this material. Please click here if you are a supplier and would like information on how to add your listing to this material.

| Physical Properties | Metric | English | Comments |
|---|---|---|---|
| Density | 8.00 g/cc | 0.289 lb/in³ | |

| Mechanical Properties | Metric | English | Comments |
|---|---|---|---|
| Hardness, Rockwell B | 79 | 79 | |
| Tensile Strength, Ultimate | 560 MPa | 81200 psi | |
| Tensile Strength, Yield | 290 MPa | 42100 psi | |
| Elongation at Break | 50 % | 50 % | in 50 mm |
| Tensile Modulus | 193 GPa | 28000 ksi | |
| Izod Impact | 150 J<br>@Temperature -195 °C | 111 ft-lb<br>@Temperature -319 °F | |
| | 150 J<br>@Temperature 21.0 °C | 111 ft-lb<br>@Temperature 69.8 °F | |
| Charpy Impact | 103 J | 76.0 ft-lb | V-notch, 30ºC |

| Electrical Properties | Metric | English | Comments |
|---|---|---|---|
| Electrical Resistivity | 0.0000740 ohm-cm | 0.0000740 ohm-cm | at 20ºC |
| Magnetic Permeability | 1.008 | 1.008 | at RT |

| Thermal Properties | Metric | English | Comments |
|---|---|---|---|
| Specific Heat Capacity | 0.500 J/g-°C<br>@Temperature 0.000 - 100 °C | 0.120 BTU/lb-°F<br>@Temperature 32.0 - 212 °F | |
| Thermal Conductivity | 14.0 - 15.9 W/m-K | 97.0 - 110 BTU-in/hr-ft²-°F | |
| Melting Point | 1375 - 1400 °C | 2507 - 2550 °F | |
| Solidus | 1375 °C | 2507 °F | |
| Liquidus | 1400 °C | 2550 °F | |
| Maximum Service Temperature, Air | 870 °C | 1600 °F | Intermittent |
| | 925 °C | 1700 °F | Continuous Service |

| Component Elements Properties | Metric | English | Comments |
|---|---|---|---|
| Carbon, C | <= 0.030 % | <= 0.030 % | |
| Chromium, Cr | 16 - 18 % | 16 - 18 % | |
| Iron, Fe | 61.9 - 72 % | 61.9 - 72 % | As Balance |
| Manganese, Mn | <= 2.0 % | <= 2.0 % | |
| Molybdenum, Mo | 2.0 - 3.0 % | 2.0 - 3.0 % | |
| Nickel, Ni | 10 - 14 % | 10 - 14 % | |
| Phosphorus, P | <= 0.045 % | <= 0.045 % | |

| | | |
|---|---|---|
| Silicon, Si | <= 1.0 % | <= 1.0 % |
| Sulfur, S | <= 0.030 % | <= 0.030 % |

**References** for this datasheet.